

Monte Maps: Real-time localization pipeline for indoor scenes

Simon Gardier Camille Trinh Lei Yang

University of Liège

{gardiersimon, maiphi.trinh, leiyang677}@gmail.com

Abstract

Location tracking systems, such as GPS, have been popular since their invention, even more now that they conveniently fit in a pocket. The tracking relies on a satellite signal and can be accurate to a few meters. However, it lacks precision indoor due to signal loss. While outdoor locations can be easily determined using images of landscapes or geographical features, indoor images often lack distinguishable features and resources. In this article, we present Monte Maps, a pipeline for real-time indoor visual localization, for which we release the code alongside localization data for the Montefiore Institute building and test data to evaluate the pipeline. Based on a sequence of images captured throughout the building, we processed it with COLMAP[11] to obtain a sparse 3D reconstruction and camera poses. Subsequently, to estimate query images position, we used MegaLoc[2] for image retrievals and camera pose estimation within the building. Monte-Maps allows the user to accurately estimate position within a few meters in the building. On our scene, Monte-Maps reaches 95.74% on localization accuracy across various challenging queries and shows real potential for a real-world application.

Code release: [monte-maps](#). Dataset release: [monte-maps](#)

1. Introduction

In the game GeoGuessr[1], the player have to find as precisely as possible a random place when they are dropped, which can be anywhere in the world. The score of the player is determined by the distance between the location they guess and the ground truth. This game is based on the use of Google Maps in outdoor settings, while Monte Maps allows you to provide a query image of a video feed inside the Montefiore building and find your location. Indoor visual localization is challenging due to repetitive corridors, low-texture environments, artificial lighting, and moving objects or persons. In the case of the Montefiore building, the corridors are similar, people might be present in the scene, and objects might be moved around. Recently, Vi-

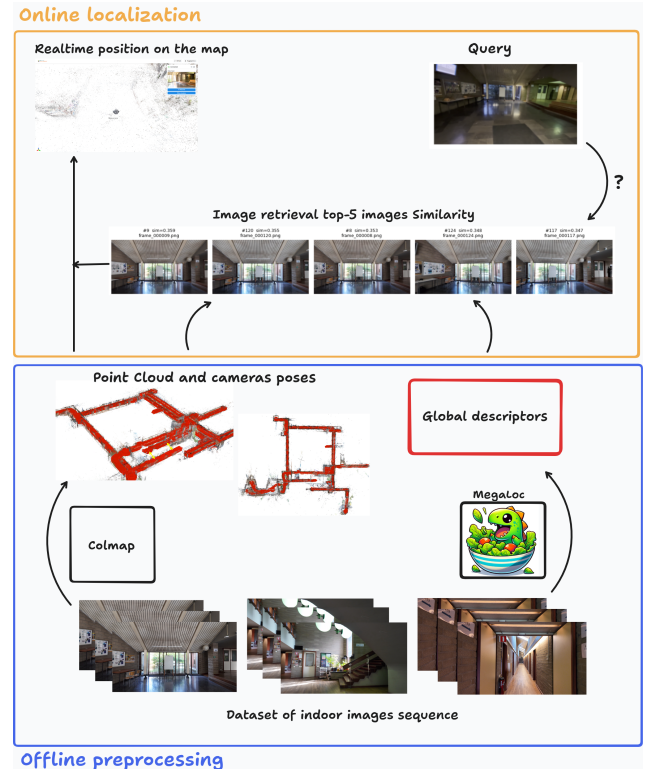


Figure 1. Monte-Maps pipeline

sual Place Recognition (VPR) methods have proposed models based on Optimal Transport Aggregation [7], solving almost completely the task at hand. Capitalizing on these models, we propose an indoor visual localization pipeline combining COLMAP for 3D reconstruction and pose estimation, MegaLoc for image retrieval and a Kalman filter for stable position estimation. This pipeline investigates the effectiveness of retrieval-based localization methods in indoor environments.

Our contributions are as follows:

- The Monte-Maps dataset: a 3D dataset of the Montefiore building, along with capture video sequence and test data for evaluation.

- Multi-threaded real-time inference pipeline for localization in the Montefiore building.
- An evaluation on the retrieval task.

2. Related Work

A pipeline for indoor visual localization involves several stages. In the case of Monte-Maps, the first stage is 3D scene reconstruction to create a 3D map of the building and estimate camera poses. Then, visual place recognition between query images and camera images is used as a position estimate.

To carry out this work, we look up the state-of-the-art in different domains: 3D reconstruction, Visual Place Recognition, and Indoor Localization.

There are two types of 3D reconstruction: optimization-based models and feed-forward (FF) models. FF models are faster than optimization models but require GPUs; they process overlapping image chunks independently and merge the resulting reconstructions. Some notable frameworks are VGGT-long, based on VGTT[4], which achieved accurate and scalable 3D reconstruction, Lingbot-map[3], based on an Autoregressive model and produces real-time dense 3D reconstruction from continuous visual streams, and Scal3r[13], based on global attention between the chunks and brought state-of-the-art pose estimation and 3D geometry accuracy.

Visual Place Recognition (VPR) aims to identify the location of a query image by matching it against a database of images. The new generation is based on self-supervised learning, seen in DinoV2Salad[7].

Indoor location is mainly used in the fields of robotics and mobile computing. Some papers, such as "Indoor localization for visually impaired travelers using computer vision on a smartphone"[5], are used in the indoor environment.

Our pipeline draws on the work of previous researchers; the two main articles are COLMAP[11] [10] and MegaLoc[2]. Feed-forward reconstruction methods were explored, such as Depth-Anything-3 streaming [8], Lingbot-map [3], VGGT-long [4], and Scal3r [13], but the results were inconclusive in our study on large-scale indoor reconstruction.

2.1. COLMAP

COLMAP is used in this study to perform 3D reconstruction of the environment based on a video sequence covering the building. COLMAP is a conventional computer vision tool based on Structure From Motion and Unstructured Multi-View Stereo. For the second stage, we did not have time to put it in place, but ideally, it would have given us a denser point cloud to produce a better mesh.

2.1.1. Structure from Motion

Structure from Motion is used to reconstruct a 3D environment from overlapping images. The initial step processes the images independently to extract local visual features from each frame. These features are then matched across multiple views to identify visual correspondences between images. Based on these correspondences, triangulation is performed to reconstruct the three-dimensional structure of the environment. Finally, optimization methods such as bundle adjustment are applied to minimize the re-projection error between the observed image features and their projected 3D positions. The objective of SFM is to generate a sparse 3D point cloud and estimate the camera positions.

2.2. MegaLoc

MegaLoc is a recent image retrieval model based on a DINOv2 backbone and a SALAD[7] aggregation network. MegaLoc is designed for visual localization and place recognition tasks.

2.2.1. SALAD

DinoV2SALAD[7] is based on a Vision Transformer architecture trained using self-supervised learning. For each image, DINOv2 divides the image into multiple patches and extracts local feature embeddings that capture semantic and visual information. SALAD (Sinkhorn Algorithm for Locally Aggregated Descriptors) then aggregates these local descriptors into a compact global representation suitable for efficient image retrieval and visual localization tasks.

2.2.2. MegaLoc

MegaLoc builds upon DINOv2 and SALAD descriptors to perform large-scale image retrieval and visual localization. While DINOv2 extracts robust visual features, MegaLoc leverages these descriptors to retrieve visually similar locations and estimate the camera position within the reconstructed environment.

The literature does not expose methods using both COLMAP and MegaLoc to solve the problem of image-based indoor localization.

3. Dataset

To tackle the localization task, in the case of Monte-Maps the first step is to capture the indoor environment of the building, as there is no pre-existing dataset. Data was acquired for both training and evaluation. The dataset is available on Hugging Face for further experimentation and potential enrichment with additional contexts. [6].

3.1. Input Data Acquisition

The video footage used to generate the point cloud has been filmed using the following specifications:

- An Osmo Pocket 3 camera with stabilization.

- Each location is filmed in two directions of walking.
- The video is sampled at 10 frames to create the input images for COLMAP.
- No obstructions or moving objects.
- All lights are lit, combined with daylight.

3.2. Input Dataset Size

The video duration is around 20 minutes with 720p image quality. The final dataset contains 12,000 images.

3.3. Evaluation dataset

The images used for the retrieval evaluation were captured using an iPhone camera and show varying lightning conditions, including darker regions. Furthermore, humans have been captured in the scene to make the retrieval harder and more realistic. This evaluation dataset contains 50 images covering the building. See the **Appendix** section for examples of test images and their retrieval results.

4. Methodology

In Monte-Maps, the pipeline is divided into two main stages: a preprocessing stage that builds the 3D environment knowledge and one that does the real-time retrieval and localization.

4.1. Preprocessing

The recorded video sequence is first decomposed into individual frames, which COLMAP then processes to perform 3D reconstruction and camera pose estimation, following the principles described in **Section 2**.

Once the reconstruction step is completed, MegaLoc is applied to extract global visual descriptors from each image and store them in a retrieval database.

4.2. Real-time retrieval and localization

Given a query image, MonteMaps computes cosine similarity scores between the query descriptor and the database descriptors to retrieve the most visually similar locations and estimate the corresponding camera position.

The last step of the pipeline is *continuous indoor tracking through temporal filtering*. While MegaLoc provides image-based localization estimates independently for each query frame, these predictions contain spatial inconsistencies caused by visual ambiguities, repetitive corridors, or dynamic objects.

To improve temporal consistency, a Kalman filter is introduced to model human walking inside the reconstructed environment. The filter uses the estimated 3D position obtained from the retrieval stage as input observations while constraining motion according to realistic walking physics dynamics.

The Kalman filter models the system state using both position and velocity components:

$$[x, y, z, v_x, v_y, v_z]$$

where (x, y, z) represents the estimated camera position and (v_x, v_y, v_z) the estimated velocity of the user.

At each time step, MegaLoc retrieves the most visually similar image from the database, and Monte-Maps recovers its associated camera pose from the COLMAP reconstruction. This position is then used as a measurement update for the Kalman filter. The prediction step estimates the next possible position according to a constant-velocity motion model, while the update step corrects the prediction using the retrieved position.

To avoid physically impossible localization jumps caused by retrieval errors, a maximum displacement threshold is introduced based on an estimated maximum human walking speed. Candidate positions exceeding this threshold are rejected, and the predicted Kalman state is retained instead.

An initialization phase is used to stabilize the system before tracking begins. During this phase, multiple localization measurements are accumulated to initialize the Kalman filter with a reliable starting position.

5. Evaluation Methodology

First stage evaluation. As no ground truth is available for the 3D reconstruction and camera poses, only a qualitative evaluation of the COLMAP results is made. **Second stage evaluation.** The second stage is evaluated on a small reference dataset that includes the ground-truth locations of the data points and the predictions it generates. In a typical assessment of images, the viewing angle is important, given that the data was captured from both directions of the walk. The only evaluation metric in our case is the reference location compared with the estimate produced by the pipeline. The main idea behind the evaluation test for this section is to feed an image into the pipeline and retrieve the top-k most similar images based on a cosine similarity score defined as follows:

$$\text{cosine similarity}(\mathbf{A}, \mathbf{B}) = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

Cosine similarity measures the angular similarity between two feature vectors. A value close to 0 indicates low similarity, whereas a value close to 1 indicates that the vectors have a highly similar orientation in the embedding space. The top-k is defined with k set to 5 for the purposes of this project. To assign a final score to the evaluation metric, the images with the highest cosine similarity are considered. Those with a distance of less than 4 meters of Euclidean distance are classified as valid, and those with a distance greater than 4 meters are classified as invalid. This score is

normalised between 0 and 1, where 1 indicates that all top-5 images are within 4 metres of each other, and 0 indicates that all images are more than 4 metres apart. This test is carried out on around fifty images to obtain a final accuracy score.

6. Experiments

6.1. Reconstruction

Our captured scene is composed of many images (12k), and the scene takes a large amount of time to process with COLMAP (between 3 and 7 days). Thus, feed-forward reconstruction methods were investigated as they provide results in a few minutes to a few hours. However, after many experiments, none of the tested methods ([12], [4], [13], [3], [8]). The methods rely on a heavy transformer predicting a 3D structure from image input. Images are split into chunks to comply with GPU memory constraints and then use traditional alignment methods to align chunks together. In our indoor scene, the alignment techniques always fail to align chunks globally. In the case of Scal3r[13], the chunks assembly generates an Out-Of-Memory error (settings: 100GB GPU, 86GB RAM, 2x48-cores CPUs), making it impossible to evaluate the final result.



Figure 2. Reconstruction failure with DA3-streaming on default settings (chunk size = 120, SIM alignment).

For the next experiments, we stuck with conventional tools like COLMAP, which gave the best results. To accelerate reconstruction, we tried the GLOMAP[9] mapper, which failed as shown in Figure 3.

The final settings used in COLMAP are the following:

- Features extraction: Shared Intrinsic, OPENCV camera model, max. 8192 features.
- Features matching: Sequential Matching with vocabulary tree (256k) and guided matching.
- Reconstruction (sparse): Incremental Mapper, default parameters

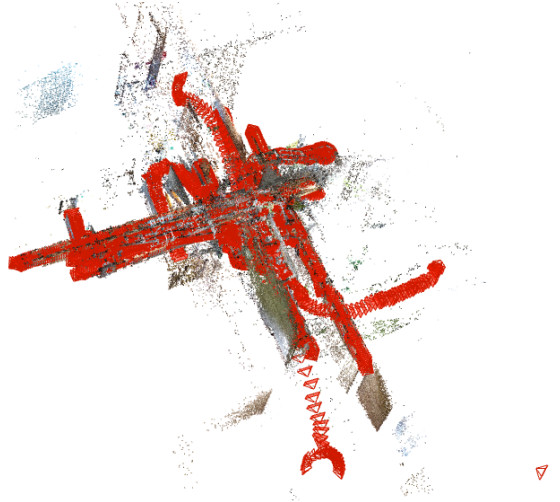


Figure 3. Reconstruction failure with COLMAP's Global Mapper[9].

The COLMAP reconstruction is close to reality, but presents the following issues:

- Drifts: As error accumulates throughout the reconstruction, the model drifts. In our case, this caused a section of the model to be duplicated. See Figure 7.
- Feature matching error: Due to errors in the features matching step of COLMAP, a small set of images has been misplaced on the map, see Figure 4.

The final result can be seen in Figure 4. As shown in Figure 5, the corridor frames provide a reasonable amount of visual clues for the reconstruction.

6.2. Retrieval and Localization

For the retrieval task, Global Descriptors are extracted for each image in the video sequence used for the reconstruction. These descriptors form a database that is loaded at inference to do real-time VPR with the query video feed images. **Qualitative results.** Qualitative results of retrieval in various conditions are shown in the Appendix. **Quantitative results.** To evaluate the retrieval stage quantitatively, we report top-1@4m average across the test query images, see Eq. 1. In our experiments, Monte-Maps reaches a top-1@4m score of **95.74%**. These results show the feasibility of indoor localization using MegaLoc. However, in rare cases (shown in the Appendix section), MegaLoc fails at identifying the correct image. Thus, a reasonable distance rejection guard rail and a simple Kalman filter have been adopted to smooth the position estimation.

$$\text{Top-k@4m}(q) = \frac{1}{k} \sum_{i=1}^k \mathbf{1}(d(r_i(q), g(q)) < 4) \quad (1)$$

With:

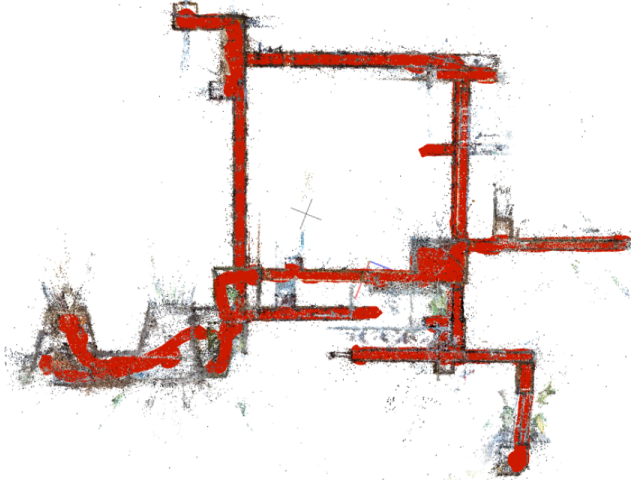


Figure 4. The final output of the COLMAP reconstruction, as seen from above. On the left is the R3 entrance, on the right is the pyramid entrance. Half of the first-floor corridor is missing due to time constraints. The drift error duplicated the main entrance staircase to the first floor and one corridor. An undesired corridor is visible in the top right, due to a failure in the features matching step.

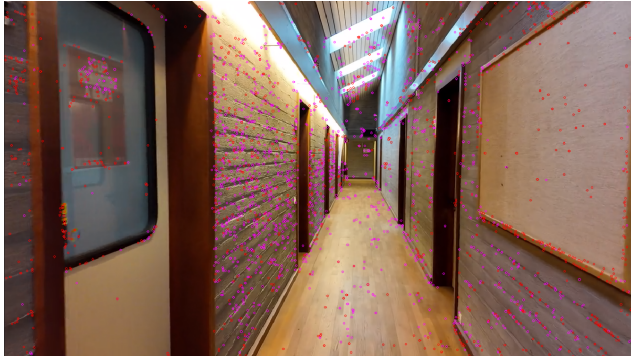


Figure 5. Extracted features from a corridor frame.

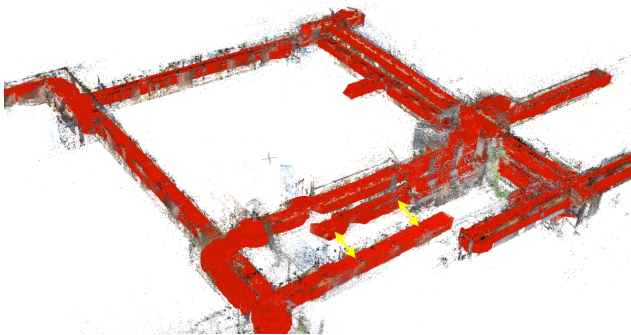


Figure 6. Case of drift in our reconstruction. The Incremental Mapper drifts, creating a disjoint corridor. See yellow arrows.

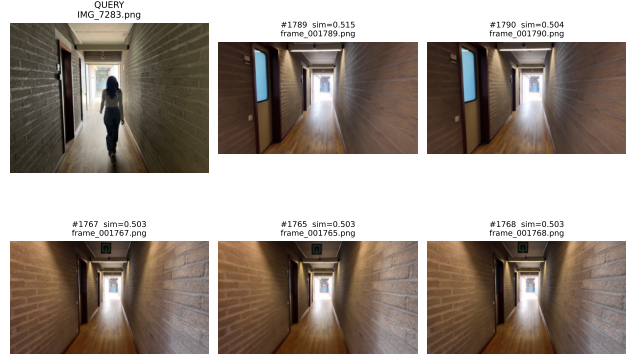


Figure 7. Top-5 MegaLoc retrieval example with obstructed scene.

- $q \in Q$: a query image
- k : number of images to retrieve
- $r_i(q)$: the i -th retrieved image for query q
- $g(q)$: ground-truth location associated with query q
- $d(\cdot, \cdot)$: spatial distance function (in meters)
- $1(\cdot)$: indicator function

6.3. Real-time inference

To provide real-time performances, we split our program into 4 components:

1. A `capture` thread, responsible for providing a new frame 10 times per second in a queue.
2. A `localization` thread, responsible for providing a new position based on frames received from the capture thread.
3. A `renderer` thread, responsible for displaying all the information. See Figure 8 for a visual
4. Finally, a main thread initializes the listed components

Our implementation supports pre-recorded videos and webcam input. Our experiments showed that the localization step takes **40ms** on average, making it possible to run localization at **25Hz**, way more than necessary to provide a smooth experience to the user. A short video demo of the inference is available at the top of our [GitHub's README](#).

7. Discussion

The results show that indoor localisation within the Montefiore framework can be implemented through our pipeline. The reconstructed point cloud enabled reliable camera pose estimation, while the retrieval stage is able to identify visually similar locations for most query images.

The difficulties encountered in this work are manifold: first, the computation time required for COLMAP, an optimisation-based method, to produce a satisfying result. A reconstruction for 12k images takes between 2 and 7 days, and multiple iterations have been necessary to reach the final results. Furthermore, it is difficult to evaluate the

3D reconstruction phase due to the lack of ground truth; the evaluation is purely qualitative. Another difficulty is the drift, where small localization and reconstruction errors accumulate progressively across the point cloud.

This study has several limitations, including the small size of the evaluation dataset for the image retrievals, which contains 50 test images. Similarly, quantitative evaluation for the Kalman filtering approach was not possible in the absence of ground truth. Furthermore, there is a lack of existing models and methods that could be used to facilitate comparison in this study.

This work opens up several possible paths for future research. These include expanding the dataset to include additional floors, lighting conditions, and dynamic scenarios, as well as exploring alternative parameters and methods for point cloud generation and retrieval methods.

8. Conclusion

This work is about an indoor visual localization pipeline combining COLMAP with MegaLoc image retrieval for localization inside the Montefiore building. We have created a dataset for this sole purpose, containing 12,000 images from a video, to generate the 3D reconstruction of the environment. The results of our experiments have demonstrated the overall effectiveness of our pipeline for indoor localization. The reconstructed map, despite having some drift that affected the accuracy, remains satisfying. Image retrieval performed reliably, succeeding in re-localizing 45 images over 47 test images. However, this approach is subject to challenges that are proper to the indoor environment, such as repetitive corridors and varying lighting settings.

Nevertheless, some work needs to be done in this area, including expanding the dataset to assess performance under various internal conditions and applying the pipeline in real time on smartphones or other devices. These are all areas for future research.

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Supplementary Material

9. Appendix

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Figure 8. The Viser[14] client for real-time localization. Processed frames are shown in the top right corner. The sparse 3D point cloud produced by COLMAP is visible in the centre. Points of interest (R7, Main entrance,...) are marked with labels.



Figure 9. An example of how image retrieval works using an image query



Figure 10. Robustness of the image retrieval despite blurry query



Figure 11. Image retrieval was a success despite varying lighting settings between the query and the dataset



Figure 12. The presence of a person in the image does not necessarily affect the prediction



Figure 13. The last retrieved image is not in the expected location



Figure 14. The query was taken in a perspective that was not filmed in the video; the database does not have the descriptor for this case

QUERY
frame_001888.png



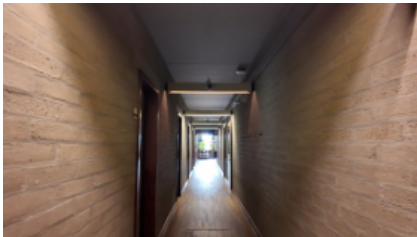
#5747 sim=0.155
frame_005747.png



#5746 sim=0.150
frame_005746.png



#5745 sim=0.146
frame_005745.png



#4685 sim=0.146
frame_004685.png



#9229 sim=0.146
frame_009229.png

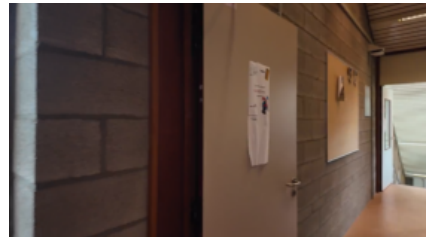


Figure 15. Localization service is having difficulty with dark queries- and is returning multiple incorrect results